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React JS Lesson 23 – React.memo & Re-render Optimization (Premium Notes)

1) What is Re-render in React

- Re-render means component runs again when state or props change.
- React updates UI after re-render.

Example:

```
setCount(count+1);
```

Explanation:

- State change triggers render again.

2) Why Performance Optimization is Needed

- Unnecessary re-renders waste performance.
- Large apps become slower.

Example:

Many child renders = extra work

Explanation:

- Optimization improves speed.

3) Parent → Child Re-render Problem

- When parent re-renders, child also re-renders even if child data unchanged.

Example:

Parent render → Child render

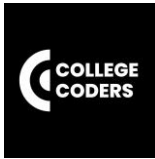
Explanation:

- Unnecessary child updates happen.

4) Introduction to React.memo

- React.memo prevents child re-render if props are unchanged.

Example:



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`export default React.memo(Child);`

Explanation:

- Child becomes memoized.

5) How React.memo Works Internally

- React compares previous props and new props.
- If same → skip render.

Example:

Old props === New props

Explanation:

- Shallow comparison happens.

6) Practical Example (Before Optimization)

- Child renders every parent update.

Example:

```
function Child(){  
  console.log("Child Render");  
  return <h2>Child</h2>  
}
```

Explanation:

- Every parent click logs child render.

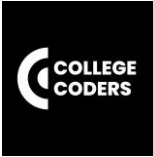
7) Practical Example (After Using React.memo)

- Child stops re-render when props same.

Example:

```
const Child = React.memo(function Child(){  
  console.log("Child Render");  
  return <h2>Child</h2>  
});
```

Explanation:



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- Logs stop if props unchanged.

8) When to Use `React.memo`

- Use when child receives same props often.
- Useful for heavy components.

Example:

Large list item components

Explanation:

- Good for repeated stable UI.

9) When NOT to Use `React.memo`

- Do not use for tiny components unnecessarily.
- Memo also has comparison cost.

Example:

Small simple components

Explanation:

- Overuse may not help.

Mini Project – Counter with Memo Child

Parent counter + memo child component

- Parent updates count.
- Child stays stable with `React.memo`.
- Observe console logs.